

Sycophancy and Stakes

Anonymous ACL submission

Abstract

Sycophancy in large language models (LLMs) is typically reported as a single rate, independent of how much the user is affected by the answer. We argue that the harm of a sycophantic response depends on the *stakes* of the prompt: the gap between the utility of an accurate response and that of a flattering one. We formalize stakes as a utility gap and test, across three domains—essay grading, factual question answering, and moral judgment—whether raising the stakes stated in a prompt reduces sycophancy. Using a pre-registered design with mixed-effects models on four models—gemini-2.5-flash, gemini-3.1-pro, gpt-oss-120b, and ministral-3-14b-instruct—, we find that stakes effects are significant but domain-dependent. On essay grading, higher stakes significantly reduce grade inflation in all four models; on factual question answering, stakes have no detectable effect in any model; and on moral judgement, the effect is mixed and negligible in magnitude. Even where stakes reduce inflation, the shift may arise as a regressive over-correction rather than an improved calibration. Our results suggest that current models do not reliably internalize prompt-stated stakes, and we push for sycophancy benchmarks to report stakes-conditioned rather than context-independent rates.

1 Introduction

“The Emperor’s New Clothes” is, in part, an allegory of sycophancy. The emperor’s “new clothes” don’t in fact exist—he is naked. His advisors, asked by the emperor how he looks, tell the emperor he looks marvelous. The emperor makes a procession in front of the whole kingdom.

For our purposes, the last part of the story is essential: at first, the advisors’ sycophancy is relatively low stakes (the emperor is in his private home), but the fact that he then marches in front

of the whole kingdom makes the situation high stakes. The effect of not telling the emperor the truth changes dramatically.

Indeed, as humans, we often weigh the costs and benefits of telling the truth. We may tell “white lies.” This form of lying has been studied extensively because it introduces a genuine tradeoff: a lie can benefit the listener, yet it runs against an aversion to lying. [Erat and Gneezy \(2012\)](#) formalize this by classifying lies according to their payoffs to the speaker and the listener, and study when people choose to tell lies that help the other person. [Biziou-van Pol et al. \(2015\)](#) likewise frame white lies as a moral conflict between lying aversion and prosocial behavior. This ties to our exploration, where the tradeoff is governed by what is at stake. [Erat and Gneezy \(2012\)](#) note that a manager deciding whether to give truthful feedback to a poorly performing employee must weigh accuracy against the risk of honest feedback reducing the employee’s confidence and future performance. This is exactly the tradeoff we study. We treat sycophancy in LLMs as an analogy to a white lie: a response that sacrifices accuracy to flatter the user.

We expand this human tendency of sycophancy to LLMs. Sycophancy in LLMs has been a well-known phenomenon for years ([Perez et al., 2023](#); [Sharma et al., 2024](#)). Previous work has made strides in evaluation benchmarks ([Cheng et al., 2025](#); [Malmqvist, 2025](#)) and the mechanistic understanding of this phenomenon. However, there is limited exploration of measuring the consequence of sycophancy in low- vs. high-stakes scenarios. Our work quantifies sycophantic responses as a function of utility. We quantify stakes as the utility of an accurate response minus the utility of a sycophantic response. Under this definition, a high-stakes user prompt carries a higher utility for an accurate response than a low-stakes prompt.

2 Related Work

Sycophancy in LLMs. Perez et al. (2023) and Sharma et al. (2024) document that RLHF models adapt their answers to a user’s stated beliefs, even against ground truth. Rrv et al. (2024) measure sycophancy using *flip rate*, the proportion of questions where a model abandons its prior answer after a user states an opinion, and break down flips into “toward user,” “away,” and “no change.” We adopt this framework for our factual-claim experiment.

Benchmarks and open-ended sycophancy. Cheng et al. (2025) introduce ELEPHANT, which uses the AITA (“Am I the Asshole?”) dataset to measure *social* sycophancy where no objective ground truth exists. Their AITA-NTA-FLIP variant pairs each prompt with a wrongdoer-perspective flip, isolating sycophancy from disagreement with a moral norm. We build directly on this dataset.

Stakes and the gap we address. Ben Natan and Tsur (2026) study sycophancy in a betting setting, where the user’s opinion is presented as a bet between two individuals—in some scenarios the user is one of the two, in others the bet is described in the third person. We extend this by varying the bet’s *magnitude*. To our knowledge, no prior work isolates stakes as an independent variable across multiple sycophancy domains. Prior benchmarks treat sycophancy as a single context-independent rate; we argue its severity depends on the response *in context*, and propose measuring sycophancy as a function of the prompt’s stakes.

3 Framework

We model the LLM response as a function of a desire to please the user and a calculation of the utility to the user. The user asks the LLM a question, and there is a set S of possible responses. Each response has a utility $u(s)$ and a flattery level $f(s)$. The LLM chooses

$$s^* = \arg \max_{s \in S} u(s) + \lambda f(s), \quad (1)$$

where $\lambda \geq 0$ controls the weight the LLM places on flattering the user. In our experiments, we modify both u and f through prompt content.

Definition 1 (Stakes). *For a fixed prompt, let $s_{\text{acc}} \in \arg \max_{s \in S} u(s)$ denote a maximally accurate response and let $s_{\text{sync}} \in \arg \max_{s \in S} f(s)$ denote a maximally sycophantic response. The*

stakes of the prompt are

$$\sigma = u(s_{\text{acc}}) - u(s_{\text{sync}}). \quad (2)$$

A high-stakes prompt has large σ ; a low-stakes prompt has small σ . Under eq. (1), increasing σ should, all else equal, push s^ toward s_{acc} .*

4 Experiments

Each experiment manipulates stakes through three levels (low/medium/high) plus baseline conditions, and tests for a stakes effect with a mixed-effects model. Essay grading and factual claim use temperature 0; ELEPHANT uses the default temperature, following the original paper. For each experiment we (i) confirm baseline sycophancy exists (a precondition; if it fails, the stakes analysis is exploratory) and (ii) fit a mixed-effects model with stakes level as an ordinal fixed effect ($\text{stakes} \in \{0, 1, 2\}$) and a per-item random intercept, testing $H_1 : \beta_{\text{stakes}} < 0$. We run all experiments on four models: gemini-2.5-flash, gemini-3.1-pro, gpt-oss-120b, and ministral-3-14b-instruct.

4.1 Essay Grading

Setup. We use the Hewlett Foundation ASAP essay dataset, restricted to the two persuasive essay sets (so a single grading rubric applies) and to essays of at least 150 words, to mitigate the known length bias of LLM evaluators (Hu et al., 2024). We sample 2,000 essays ($n = 1,996$ after model refusals). The model grades each essay on an A–F scale under five conditions: a knowledge baseline (KB, no user signal), a user baseline (“I wrote this essay.”), and three stakes levels in which the user additionally states the assignment’s grade weight—low (10%), medium (25%), high (90%).

Metric. We encode grades as $F=0, \dots, A=4$. For essay e and condition c , *grade inflation* is $\Delta_c(e) = g_c(e) - g_{\text{kb}}(e)$, the signed shift from the model’s own KB grade. Positive is inflation, negative deflation. The precondition tests whether the user baseline inflates grades ($\Delta_{\text{user}} > 0$); the trend test fits $\Delta_c(e) \sim \beta_{\text{stakes}} \cdot \text{stakes} + (1 \mid e)$.

4.2 Factual Claim

Setup. We sample 2,000 questions from MMLU-Pro (Wang et al., 2024). Extending Ben Natan and Tsur (2026), the user states an answer in the context of a bet whose magnitude encodes stakes: \$5 (low), \$100 (medium), \$200 (high). We run

each stakes level with the user holding an incorrect answer (the sycophancy test) and, as a control, the correct answer.

Metric. Following Rrv et al. (2024), the primary metric is the flip-to-user rate: the proportion of questions where the model abandons its KB answer and adopts the user’s stated answer. The trend test is a logistic mixed-effects model $\text{flip}_{\rightarrow u} \sim \beta_{\text{stakes}} \cdot \text{stakes} + (1 \mid q)$. For baseline, the user holds the incorrect answer; for stakes, we run both.

4.3 Moral Sycophancy (ELEPHANT)

Setup. We use the AITA-NTA-FLIP dataset (Cheng et al., 2025): 1,591 prompt pairs, each an original post (crowd consensus NTA) and a flipped, wrongdoer-perspective version (consensus YTA). We define stakes as the user’s stated *actionability*: low (“I’m processing this for myself; no action planned.”), medium (“I’m planning to talk to them about this next week.”), high (“I’m planning to end the relationship tonight based on what you say.”).

Metric. The primary metric is moral sycophancy: the rate at which the model labels *both* perspectives NTA, validating both sides of a conflict with a clear wrongdoer. The trend test is a logistic mixed-effects model $\text{moral_sys} \sim \beta_{\text{stakes}} \cdot \text{stakes} + (1 \mid p)$.

5 Results

Table 1 summarizes all three experiments across the four models.

Essay grading: stakes reduce inflation in every model. The essay stakes effect replicates across all four models: β_{stakes} is significantly negative in every case (gemini-2.5-flash -0.136 , gemini-3.1-pro -0.068 , gpt-oss-120b -0.038 , minstral-3-14b-instruct -0.066 ; all one-sided $p < 10^{-13}$). The models differ, however, in their knowledge and user baseline. gpt-oss-120b and minstral-3-14b-instruct inflate at the user baseline ($\Delta_{\text{user}} = +0.148$, $d = 0.27$; and $+0.341$, $d = 0.49$), and stakes move inflation toward zero. gemini-2.5-flash and gemini-3.1-pro instead start at or below zero ($\Delta_{\text{user}} = +0.076$, $d = 0.15$; and -0.044 , $d = -0.12$), and stakes further increase grade deflation ($-0.72 \rightarrow -0.99$ and $-0.30 \rightarrow -0.43$). Because both Gemini models grade below their own KB level in response to stakes, we interpret this as over-correction toward harshness, not

improved calibration. For gemini-3.1-pro the precondition fails ($d = -0.12$: the user signal deflates rather than inflates), so we mark this as exploratory. The stakes effect on essay grading is thus robust in direction (decreases with stakes), but model-dependent in behavior: either reduces inflation or over-corrects into deflation.

Factual claim: stakes have no detectable effect. No model shows a stakes effect on factual sycophancy. Baseline sycophancy is present in all four (flip-to-user at the user baseline 0.218/0.018/0.039/0.344; Cohen’s d from 0.13 to 0.72), but across stakes the flip-to-user rate is flat for every model (high-minus-low changes of -0.008 , $+0.002$, -0.006 , $+0.004$). Factual sycophancy is insensitive to the bet magnitude.

Moral sycophancy: mixed, with a medium-stakes peak in two models. Baseline moral sycophancy is significant in every model and varies widely across models (both-NTA at baseline 0.343/0.122/0.524/0.679; d from 0.37 to 1.45). The stakes effect, however, is small and inconsistent. gpt-oss-120b shows a clear decline in stakes *low* $\rightarrow$ *medium* $\rightarrow$ *high* (-0.030 , $0.513 \rightarrow 0.492 \rightarrow 0.483$). gemini-3.1-pro declines marginally (-0.008 , $0.116 \rightarrow 0.113 \rightarrow 0.108$). gemini-2.5-flash and minstral-3-14b-instruct move in the *opposite* direction ($+0.041$ and $+0.059$): for these models, increasing stakes slightly *increases* moral sycophancy. In both cases this is driven by a non-monotone medium-stakes peak rather than a steady trend—gemini-2.5-flash runs $0.307 \rightarrow 0.382 \rightarrow 0.326$ across low/medium/high and minstral-3-14b-instruct runs $0.752 \rightarrow 0.847 \rightarrow 0.772$. No consistent stakes effect on moral judgment emerges.

Conclusion. We formalized stakes as a utility gap between accuracy and sycophancy and tested, across three domains and two models, whether higher-stakes prompts reduce sycophancy. Stakes effect is present but is domain-dependent. On essay grading it replicates cleanly: all four models reduce grade inflation as stakes rise, though Gemini models over-correct into regressive grade deflation. On factual question answering, stakes effect is absent: no model’s flip-to-user rate responds to the bet magnitude. On moral judgment, it is mixed and small: gpt-oss-120b declines substantially, gemini-3.1-pro negligibly, and gemini-

Model	User baseline	Low	Medium	High	β_{stakes}
<i>Essay grading</i> — grade inflation Δ_c vs. KB (grade units; + inflation, − deflation)					
gemini-2.5-flash	+0.076 (.05, .10)	−0.720 (−.76, −.69)	−0.726 (−.76, −.69)	−0.992 (−1.03, −.95)	−0.136
gemini-3.1-pro [†]	−0.044 (−.06, −.03)	−0.296 (−.32, −.28)	−0.326 (−.35, −.31)	−0.430 (−.45, −.41)	−0.068
gpt-oss-120b	+0.148 (.12, .17)	+0.138 (.11, .16)	+0.131 (.11, .16)	+0.063 (.04, .09)	−0.038
ministral-3-14b-instruct	+0.341 (.31, .37)	+0.323 (.29, .35)	+0.299 (.27, .33)	+0.191 (.16, .22)	−0.066
<i>Factual claim</i> — flip-to-user rate (user holds incorrect answer)					
gemini-2.5-flash	0.218 (.20, .24)	0.155 (.14, .17)	0.154 (.14, .17)	0.147 (.13, .16)	−0.029
gemini-3.1-pro	0.018 (.01, .03)	0.010 (.01, .01)	0.011 (.01, .02)	0.012 (.01, .02)	+0.071
gpt-oss-120b	0.039 (.03, .05)	0.033 (.03, .04)	0.031 (.02, .04)	0.027 (.02, .03)	−0.111
ministral-3-14b-instruct	0.344 (.32, .37)	0.137 (.12, .15)	0.142 (.13, .16)	0.141 (.13, .16)	+0.019
<i>Moral sycophancy</i> — both-NTA rate					
gemini-2.5-flash	0.345 (.32, .37)	0.307 (.29, .33)	0.382 (.36, .41)	0.326 (.30, .35)	+0.041
gemini-3.1-pro	0.122 (.11, .14)	0.116 (.10, .13)	0.113 (.10, .13)	0.108 (.09, .12)	−0.040
gpt-oss-120b	0.524 (.50, .55)	0.513 (.49, .54)	0.492 (.47, .52)	0.483 (.46, .51)	−0.146
ministral-3-14b-instruct	0.679 (.66, .70)	0.752 (.73, .77)	0.847 (.83, .87)	0.772 (.75, .79)	+0.059

Table 1: Primary metric by condition for all four models, with 95% CIs in parentheses ($n = 2,000$ per cell; moral 1,591 pairs). β_{stakes} is the stakes-trend coefficient ($H_1: \beta_{\text{stakes}} < 0$); grade units per step for essay, log-odds per step for factual and moral. [†] For gemini-3.1-pro the essay precondition fails ($\Delta_{\text{user}} = -0.044$, $d = -0.12$); its essay stakes analysis is exploratory. Essay KB mean grade 1.37/1.71/0.77/1.66; factual KB accuracy 0.688/0.875/0.760/0.508.

2.5-flash and ministral-3-14b-instruct move against the hypothesis with a medium-stakes peak. Sycophancy reduction in response to stakes is therefore neither constant across domains nor consistent in magnitude—evidence that current models do not reliably internalize prompt-stated stakes. This further confirms our claim that single context-independent sycophancy rates and benchmarks may misestimate harm in higher-stakes domains.

Limitations

Our experiments use four models (gemini-2.5-flash, gemini-3.1-pro, gpt-oss-120b, and ministral-3-14b-instruct); the results may not transfer to other families. We use only black-box testing to evaluate whether models internalize stakes; future work could use mechanistic interpretability to examine model activations under the prompt changes. We use one prompt per stakes level per experiment, which may not account for model sensitivity to specific prompt wordings—this is especially relevant to the medium-stakes peak we observe in the moral-judgment experiment, which we cannot separate from the specific medium-stakes prompt. Our evaluation focuses on single output responses; future work could examine probability distributions to measure probability shifts across stakes conditions (Atwell et al., 2025). Finally, future work could build context-aware sycophancy

benchmarks that report stakes-conditioned rates.

References

- Katherine Atwell, Pedram Heydari, Anthony Sicilia, and Malihe Alikhani. 2025. [BASIL: Bayesian assessment of sycophancy in LLMs](#). *arXiv preprint arXiv:2508.16846*.
- Shahar Ben Natan and Oren Tsur. 2026. [Not your typical sycophant: The elusive nature of sycophancy in large language models](#). *arXiv preprint arXiv:2601.15436*.
- Laura Bizziou-van Pol, Jana Haenen, Arianna Novaro, Andrés Occhipinti Liberman, and Valerio Capraro. 2015. Does telling white lies signal pro-social preferences? *Judgment and Decision Making*, 10(6):538–548.
- Myra Cheng, Sunny Yu, Cinoo Lee, Pranav Khadpe, Lujain Ibrahim, and Dan Jurafsky. 2025. [ELEPHANT: A benchmark for measuring social sycophancy in language models](#). *arXiv preprint arXiv:2505.13995*.
- Sanjiv Erat and Uri Gneezy. 2012. White lies. *Management Science*, 58(4):723–733.
- Zhengyu Hu, Linxin Song, Jieyu Zhang, and 1 others. 2024. [Explaining length bias in LLM-based preference evaluations](#). *arXiv preprint arXiv:2407.01085*.
- Lars Malmqvist. 2025. [Sycophancy in large language models: Causes and mitigations](#). *arXiv preprint arXiv:2411.15287*.
- Ethan Perez, Sam Ringer, Kamile Lukosiute, and 1 others. 2023. Discovering language model behaviors with model-written evaluations. In *Findings of the Association for Computational Linguistics: ACL 2023*.

Aswin Rrv, Nemika Tyagi, Md Nayem Uddin, Neeraj Varshney, and Chitta Baral. 2024. [Chaos with keywords: Exposing large language models’ sycophantic behavior](#). *arXiv preprint arXiv:2406.15163*.

Mrinank Sharma, Meg Tong, Tomasz Korbak, and 1 others. 2024. Towards understanding sycophancy in language models. In *The Twelfth International Conference on Learning Representations*.

Yubo Wang, Xueguang Ma, Ge Zhang, and 1 others. 2024. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. In *Advances in Neural Information Processing Systems (NeurIPS)*.

A Appendix